

MT1008 Multivariate Calculus

Assignment-2

Question 1

- (a) Let $g(x, y) = x^2 \ln(x + y)$.
- (i) Evaluate $g(3, 1)$.
 - (ii) Find and sketch the domain of g .
 - (iii) Find the range of g .
- (b) Find and sketch the domain of the function:

$$h(x, y) = \ln(x^2 + y^2 - 9)$$

Question 2

The temperature-humidity index I (or humidex, for short) is the perceived air temperature when the actual temperature is T and the relative humidity is h , so we can write $I = f(T, h)$. The following table of values of I is an excerpt from a table compiled by the National Oceanic & Atmospheric Administration.

Table 1: Apparent temperature as a function of temperature and humidity

Relative humidity (%)	20	30	40	50	60	70
T						
20	20	20	20	21	22	23
25	25	25	26	28	30	32
30	30	31	34	36	38	41
35	36	39	42	45	48	51
40	43	47	51	55	59	63

- (a) What is the value of $f(35, 70)$? What is its meaning?
- (b) For what value of h is $f(40, h) = 55$?
- (c) For what value of T is $f(T, 50) = 42$?
- (d) What are the meanings of the functions $I = f(25, h)$ and $I = f(35, h)$? Compare the behavior of these two functions of h .

Question 3

(a) Evaluate (if it exists)

$$\lim_{(x,y) \rightarrow (-2,3)} \frac{x^2y + 1}{x^3y^2 - 2x}$$

(b) Show that the following limit does not exist.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{x^2 + 3y^2}$$

Question 4

The wave heights h in the open sea depend on the speed v of the wind and the length of time t that the wind has been blowing at that speed. Values of the function $h = f(v, t)$ are recorded in feet in the following table.

Table 2: Wave heights as a function of wind speed and duration

Duration (hours)	5	10	15	20	30	40	50
v							
20	0.6	0.6	0.6	0.6	0.5	0.6	0.6
30	1.2	1.3	1.5	1.5	1.5	1.6	1.6
40	1.5	2.2	2.4	2.5	2.7	2.8	2.8
60	2.8	4.0	4.9	5.2	5.5	5.8	5.9
80	4.3	6.4	7.7	8.6	9.5	10.1	10.2
100	5.8	8.9	11.0	12.2	13.8	14.7	15.3
120	7.4	11.3	14.4	16.6	19.0	20.5	21.1

(a) What are the meanings of the partial derivatives $\partial h / \partial v$ and $\partial h / \partial t$?

(b) Estimate the values of $f_v(40, 15)$ and $f_t(40, 15)$. What are the practical interpretations of these values?

(c) What appears to be the value of the following limit?

$$\lim_{t \rightarrow \infty} \frac{\partial h}{\partial t}$$

Question 5

- (a) Find the indicated partial derivative(s).

$$f(x, y) = x^4y^2 - x^3y; \quad f_{xxx}, \quad f_{yx}$$

- (b) Show that $f(x, y) = xe^{xy}$ is differentiable at $(1, 0)$ and find its linearization there. Then use it to approximate $f(1.1, -0.1)$.

- (c) Find an equation of the tangent plane to the given surface at the specified point.

$$z = e^{x-y}, \quad (2, 2, 1)$$

Question 6

- (a) If $z = x^2y + 3xy^4$, where $x = \sin 2t$ and $y = \cos t$, find dz/dt when $t = 0$.
- (b) If $z = e^x \sin y$, where $x = st^2$ and $y = s^2t$, find $\partial z/\partial s$ and $\partial z/\partial t$.
- (c) If $u = x^4y + y^2z^3$, where $x = rse^t$, $y = rs^2e^{-t}$, and $z = r^2s \sin t$, find the value of $\partial u/\partial s$ when $r = 2$, $s = 1$, $t = 0$.
- (d) Use the Chain Rule to find the indicated partial derivatives when $r = 2$, $\theta = \pi/2$:

$$w = xy + yz + zx, \quad x = r \cos \theta, \quad y = r \sin \theta, \quad z = r\theta;$$

$$\frac{\partial w}{\partial r}, \quad \frac{\partial w}{\partial \theta}$$

Question 7

- (a) Find the directional derivative of the function $f(x, y) = x^2y^3 - 4y$ at the point $(2, -1)$ in the direction of the vector $\mathbf{v} = 2\mathbf{i} + 5\mathbf{j}$.
- (b) Find the tangent plane to the surface $z = 2x^2 + y^2$ at the point $(1, 1, 3)$.

Question 8

- (a) Find the gradient of f .
- (b) Evaluate the gradient at the point P .
- (c) Find the rate of change of f at P in the direction of the vector u .

$$f(x, y, z) = y^2e^{xyz}, \quad P(0, 1, -1), \quad u = \left\langle \frac{3}{13}, \frac{4}{13}, \frac{12}{13} \right\rangle$$

Question 9

- (a) Find the local maximum and minimum values and saddle points of

$$f(x, y) = x^4 + y^4 - 4xy + 1.$$

- (b) Find the local maximum and minimum values and saddle point(s) of

$$f(x, y) = xy - 2x - 2y - x^2 - y^2.$$

You are encouraged to use a calculator or computer to graph the function with a domain and viewpoint that reveals all the important aspects of the function.

Question 10

Compute the local minima of the given function by using gradient descent algorithm by taking step size as 0.15 and initial point as $(2, 2)$. Perform three iterations.

$$f(x, y) = 4x^2 + 2.5y^2 + 3xy - 5.5x - 4.1y$$